

Properties of Congruence ($\cong$)

P.101

1. **Reflexive Property of Congruence**

$$\overline{AB} \cong \overline{AB}$$

$$\angle A \cong \angle A$$

$$\overline{AB} \cong \overline{BA}$$

2. **Symmetric Property of Congruence**

$$\text{If } \overline{AB} \cong \overline{CD}, \text{ then } \overline{CD} \cong \overline{AB}.$$

$$\text{If } \angle A \cong \angle B, \text{ then } \angle B \cong \angle A.$$

3. **Transitive Property of Congruence**

$$\text{If } \overline{AB} \cong \overline{CD} \text{ and } \overline{CD} \cong \overline{RT},$$

$$\text{then } \overline{AB} \cong \overline{RT}. \text{ (same with angles)}$$

Definition of Congruent Segments (or Angles)

$$\text{If } \overline{AB} \cong \overline{CD}, \text{ then } AB = CD.$$

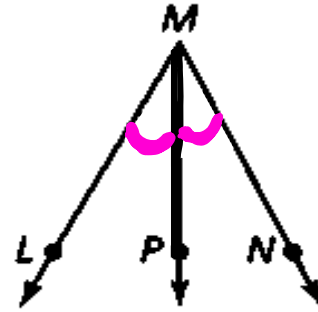
$$\text{If } RT = HM, \text{ then } \overline{RT} \cong \overline{HM}.$$

Example:

$\angle A \cong \angle B$		given
$m\angle A = m\angle B$		$\text{Def. of } \cong \text{ Angles}$

Given: $\overrightarrow{MP}$ bisects $\angle LMN$

Prove: $2(m\angle LMP) = m\angle LMN$



Statements	Reasons
1. $\overrightarrow{MP}$ bisects $\angle LMN$	1. given
2. $\angle LMP \cong \angle NMP$	2. Def. of angle bisector
3. $m\angle LMP = m\angle NMP$	3. Def. of $\cong$ angles
4. $m\angle LMP + m\angle NMP = m\angle LMN$	4. Angle Addition Postulate
5. $m\angle LMP + m\angle LMP = m\angle LMN$	5. Substitution Prop.
6. $2(m\angle LMP) = m\angle LMN$	6. Distributive Prop.